

CampusEvents

Campus Event & Ticket Booking System

COP 4710 — Database Systems University of South Florida Spring 2026

Final Report — Due May 1, 2026

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1. Team Roster

This report is the final submission for the CampusEvents term project. It is submitted by the two-person team listed below. Per Dr. Tu's course guidelines, two-person groups carry the same deliverable expectations as a three-person group, and the scope of CampusEvents reflects that.

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2. System Overview

CampusEvents is a web-based enterprise information system for managing campus events at a university. The platform serves three distinct user roles: **students**, who browse events and register for them; **organizers**, who create and manage them; and **administrators**, who control the entire system — users, events, and categories. The core design goal is a centralized, queryable system that replaces ad-hoc flyers and scattered email announcements.

The system follows a standard three-tier architecture. The **interface tier** is a React 19 single-page application built with Vite, styled with Tailwind CSS v4. The **application tier** is a TypeScript REST API running on the Bun runtime via the Hono framework. The **data tier** is a PostgreSQL 16 relational database, accessed through the `postgres.js` driver using raw parameterized SQL — no ORM is used at any layer, per course requirements.

Tier	Technology	Port	Responsibility
Interface	React 19 + Vite + Tailwind CSS v4	:5173	Rendering, routing, client-side state
Application	Bun + Hono (TypeScript REST API)	:3000	Validation, auth, SQL via <code>postgres.js</code>
Data	PostgreSQL 16 (Docker)	:5432	Storage, constraints, referential integrity

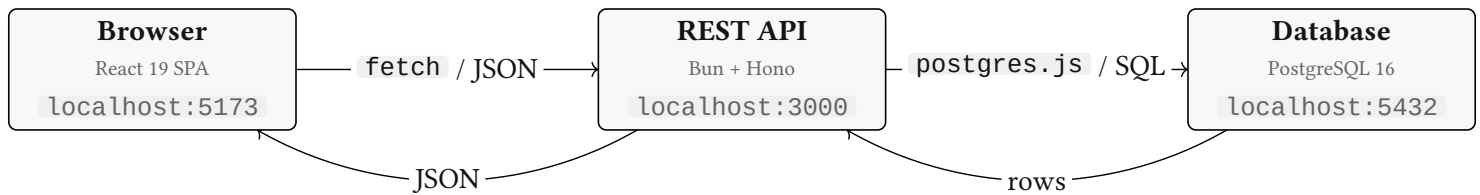


Figure 1: Three-tier architecture — the React SPA issues `fetch` calls to the Hono REST API, which sends parameterized SQL to PostgreSQL via the `postgres.js` driver.

`postgres.js` serves as the JDBC/ODBC equivalent in this stack: a lightweight PostgreSQL client for JavaScript that sends parameterized SQL directly to the database using tagged template literals (e.g., `sql SELECT * FROM users WHERE id = ${id}`). All user-supplied values are automatically escaped, preventing SQL injection at the driver level.

3. Database Design

3.1. ER Diagram

The entity-relationship diagram on the next page represents the conceptual data model for CampusEvents in **Chen notation**. Strong entities are drawn as single rectangles, weak entities as double rectangles; relationships are diamonds (double diamonds when identifying); attributes are ellipses, with key attributes underlined and composite attributes shown as nested ellipses. Edge style indicates participation: a single line is partial, a double line is total. Cardinality ratios (1, N, M) are labelled directly on each edge. Every column from `schema.sql` is represented; SQL data types and constraints are documented in the **Tables & Attributes** section that follows.

`TICKETS` is modelled as a **weak entity** because every ticket's existence is bound to both its owning user and its event — the schema enforces this via `ON DELETE CASCADE` on both foreign keys, and the natural identifier of a ticket is the (user, event) pair. The Chen diagram captures this by giving `TICKETS` a partial key (`confirmation_code`) and connecting it to `USERS` and `EVENTS` through identifying relationships. The `event_categories` SQL bridge table is correspondingly resolved into the M:N `categorized-as` relationship rather than a standalone entity, since it carries no attributes of its own.

The `IMAGES` entity persists uploaded image binaries (avatars and event banners) so that media survive container restarts and reseeds. The physical schema stores these as URL strings in `USERS.profile_photo` and `EVENTS.banner_url` rather than enforced foreign-key columns, but the conceptual relationships are real and are modelled at the Chen level as `has-avatar` and `has-banner` (both 0:1 from the parent side). The diagram is rotated onto a landscape page so that all attribute ellipses remain legible at print size.

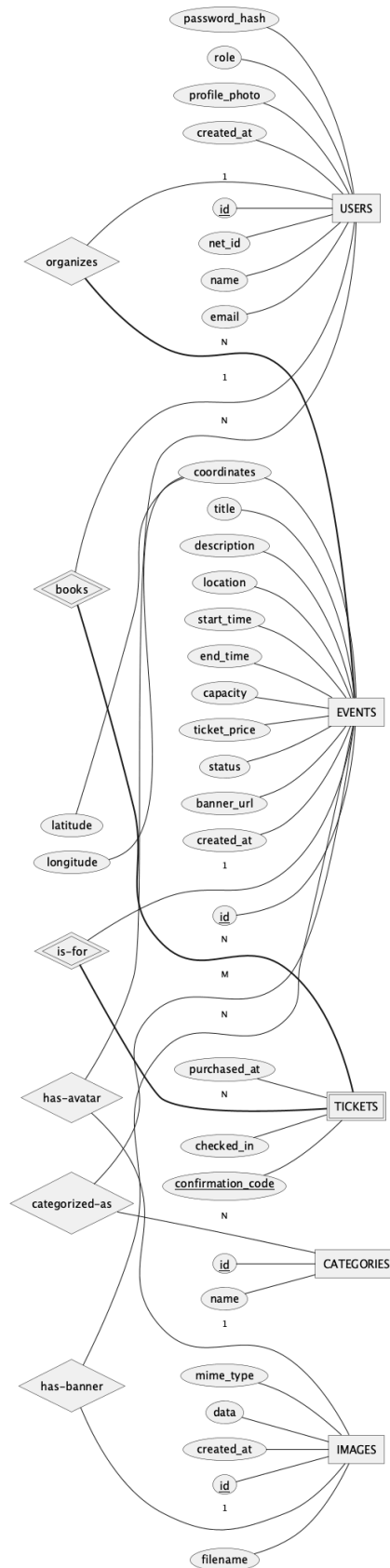


Figure 2: ER diagram for CampusEvents. Strong entities: **USERS**, **EVENTS**, **CATEGORIES**, **IMAGES**. Weak entity: **TICKETS** (identified through **USERS** and **EVENTS** via **books** and **is-for**). Relationships: **organizes** (**USERS**→**EVENTS**, 1:N, total on **EVENTS**); **books**, **is-for** (identifying, 1:N, total on **TICKETS**); **categorized-as** (**EVENTS**↔**CATEGORIES**, M:N); **has-avatar** (**USERS**→**IMAGES**, N:1), **has-banner** (**EVENTS**→**IMAGES**, N:1) — the latter two capture URL-based references that exist conceptually even though the physical schema does not enforce them with foreign keys.

3.2. Tables & Attributes

USERS — Represents every account in the system. The `role` enum determines what the account can do: `student` accounts browse and register; `organizer` accounts create and manage events; `admin` accounts have full system access including role assignment and category management. Passwords are stored exclusively as bcrypt hashes; plaintext is never persisted.

Attribute	Type	Constraint	Notes
<code>id</code>	<code>SERIAL</code>	<code>PRIMARY KEY</code>	Auto-incremented surrogate key
<code>net_id</code>	<code>VARCHAR(50)</code>	<code>NOT NULL, UNIQUE</code>	University NetID (e.g., <code>jsmith22</code>)
<code>name</code>	<code>VARCHAR(100)</code>	<code>NOT NULL</code>	Full display name
<code>email</code>	<code>VARCHAR(150)</code>	<code>NOT NULL, UNIQUE</code>	University email address
<code>password_hash</code>	<code>TEXT</code>	<code>NOT NULL</code>	bcrypt hash; plaintext never stored
<code>role</code>	<code>user_role</code>	<code>DEFAULT 'student'</code>	<code>admin</code> <code>organizer</code> <code>student</code>
<code>profile_photo</code>	<code>TEXT</code>	nullable	URL path to uploaded avatar
<code>created_at</code>	<code>TIMESTAMP</code>	<code>DEFAULT NOW()</code>	Account creation timestamp

EVENTS — The central entity of the system. Every event has a designated organizer (FK to `USERS`), a time range, a physical venue, and a capacity ceiling. The `status` enum drives filtering across the UI; its value is computed at query time from the event's dates rather than stored statically, so completed events update automatically without a background job. Two additional fields, `latitude` and `longitude`, store GPS coordinates for the Mapbox map preview.

Attribute	Type	Constraint	Notes
<code>id</code>	<code>SERIAL</code>	<code>PRIMARY KEY</code>	Auto-incremented surrogate key
<code>title</code>	<code>VARCHAR(200)</code>	<code>NOT NULL</code>	Event display name
<code>description</code>	<code>TEXT</code>	nullable	Optional long-form description
<code>location</code>	<code>VARCHAR(200)</code>	<code>NOT NULL</code>	Venue name or address string
<code>start_time</code>	<code>TIMESTAMP</code>	<code>NOT NULL</code>	Event start date and time

Attribute	Type	Constraint	Notes
end_time	TIMESTAMP	NOT NULL	CHECK enforces end_time > start_time
capacity	INTEGER	CHECK > 0	Maximum attendee count
ticket_price	NUMERIC(10,2)	DEFAULT 0.00, CHECK >= 0	0.00 denotes a free event
status	event_status	DEFAULT 'upcoming'	upcoming ongoing completed cancelled
organizer_id	INTEGER	FK → USERS.id	References the event creator
banner_url	TEXT	nullable	URL path to uploaded banner image
latitude	DOUBLE PRECISION	nullable	GPS latitude for map pin
longitude	DOUBLE PRECISION	nullable	GPS longitude for map pin
created_at	TIMESTAMP	DEFAULT NOW()	Record insertion time-stamp

TICKETS — A ticket represents a confirmed booking between a user and an event. The composite `UNIQUE(user_id, event_id)` constraint enforces the one-ticket-per-user-per-event business rule at the schema level — if a student tries to purchase a second ticket for the same event, PostgreSQL raises error `23505` and the API returns `409 Conflict`. Each ticket carries a short alphanumeric `confirmation_code` displayed to the user and used for check-in.

Attribute	Type	Constraint	Notes
<u>id</u>	SERIAL	PRIMARY KEY	Auto-incremented surrogate key
user_id	INTEGER	FK → USERS.id	References the ticket holder
event_id	INTEGER	FK → EVENTS.id	References the booked event
purchased_at	TIMESTAMP	DEFAULT NOW()	Booking timestamp
checked_in	BOOLEAN	DEFAULT FALSE	Toggled TRUE on event entry

Attribute	Type	Constraint	Notes
confirmation_code	VARCHAR(20)	UNIQUE	Short alphanumeric booking code
(user_id, event_id)	—	UNIQUE composite	Prevents duplicate bookings

CATEGORIES — A lookup table of event tags. The schema seeds seven default categories (Academic, Music, Sports, Career, Social, Arts, Technology) using `ON CONFLICT DO NOTHING`, making the file idempotent across restarts. Admins may add categories at runtime.

EVENT_CATEGORIES — Resolves the many-to-many relationship between **EVENTS** and **CATEGORIES**. The composite primary key (event_id, category_id) ensures each pairing is unique, and `ON DELETE CASCADE` on both foreign keys keeps the join table clean automatically.

IMAGES — Stores uploaded event banners and user profile photos as base-64 data directly in the database rather than on the file system. This design choice means media survives container restarts, branch switches, and fresh clones without any external file-server configuration.

Attribute	Type	Constraint	Notes
id	SERIAL	PRIMARY KEY	
filename	TEXT	NOT NULL, UNIQUE	Generated name used as the URL key
mime_type	TEXT	NOT NULL	image/jpeg, image/png, etc.
data	TEXT	NOT NULL	Base-64-encoded binary data
created_at	TIMESTAMP	DEFAULT NOW()	

3.3. Relationship Sets

Entity A	Entity B	Name	Cardinality	Description
USERS	EVENTS	<i>organizes</i>	1 : N	An organizer may create many events; each event has exactly one organizer
USERS	TICKETS	<i>books</i>	1 : N	A student may hold tickets to many events; each ticket belongs to one user
EVENTS	TICKETS	<i>has</i>	1 : N	An event may have many tickets; each ticket corresponds to exactly one event

Entity A	Entity B	Name	Cardinality	Description
EVENTS	CATEGORIES	<i>tagged with</i>	M : N	Resolved via EVENT_CATEGORIES; an event may belong to multiple categories

3.4. Constraints & Indexes

The following business rules are enforced at the schema level, independent of application behavior:

- `CHECK (end_time > start_time)` — Prevents events from being inserted with an inverted time range.
- `CHECK (capacity > 0)` — Every event must allow at least one attendee.
- `CHECK (ticket_price >= 0)` — Prevents negative ticket prices.
- `UNIQUE (user_id, event_id)` on TICKETS — One ticket per user per event; violation raises SQL state 23505.
- `UNIQUE (organizer_id, title, start_time)` on EVENTS — Prevents duplicate events from the same organizer at the same time; also the target of seed data's `ON CONFLICT` clause.
- `ON DELETE CASCADE` on TICKETS and EVENT_CATEGORIES — Deleting a user or event automatically removes all dependent ticket and category-link rows.

Five indexes support the most common query patterns:

Index	Column(s)	Query it accelerates
<code>idx_events_status</code>	<code>events(status)</code>	Filter upcoming / ongoing events on every page load
<code>idx_events_organizer</code>	<code>events(organizer_id)</code>	Organizer dashboard event list
<code>idx_events_start_time</code>	<code>events(start_time)</code>	Date-range filtering (Q10 <code>BETWEEN</code>)
<code>idx_tickets_user</code>	<code>tickets(user_id)</code>	My Tickets page — Q4 multi-JOIN
<code>idx_tickets_event</code>	<code>tickets(event_id)</code>	Attendee list per event, capacity sub-query

3.5. Views

Two SQL views are defined in `schema.sql` using `CREATE OR REPLACE VIEW`, making them idempotent across every backend restart. They provide role-scoped projections of the underlying tables — a public-facing browse surface and an organizer-facing aggregation.

`v_upcoming_events` — Joins EVENTS with USERS and pre-filters to `status = 'upcoming'`. Exposes organizer name and email alongside event fields while hiding internal columns such as `organizer_id` and `created_at` that are irrelevant to a browsing student.

v_event_stats — Aggregates ticket counts per event using `GROUP BY` and `COUNT(*)` over the `EVENTS-TICKETS` join. This view is queried directly by `GET /api/events/stats`, which powers the organizer dashboard’s stat cards.

```
-- v_upcoming_events: public browse projection (student-facing)
CREATE OR REPLACE VIEW v_upcoming_events AS
    SELECT e.id, e.title, e.description, e.location,
           e.start_time, e.end_time, e.capacity, e.ticket_price,
           e.status, e.banner_url, e.latitude, e.longitude,
           u.name AS organizer_name,
           u.email AS organizer_email
    FROM events e
    INNER JOIN users u ON e.organizer_id = u.id
    WHERE e.status = 'upcoming'
    ORDER BY e.start_time;

-- v_event_stats: organizer ticket-sales summary (organizer-facing)
CREATE OR REPLACE VIEW v_event_stats AS
    SELECT e.id AS event_id,
           e.title, e.status, e.capacity,
           e.ticket_price, e.end_time, e.organizer_id,
           u.name AS organizer_name,
           COUNT(t.id)::int AS tickets_sold,
           (e.capacity - COUNT(t.id)::int) AS spots_remaining
    FROM events e
    INNER JOIN users u ON e.organizer_id = u.id
    LEFT JOIN tickets t ON e.id = t.event_id
    GROUP BY e.id, e.title, e.status, e.capacity,
             e.ticket_price, e.end_time, e.organizer_id, u.name;
```

The `GET /api/events/stats` route queries `v_event_stats` directly:

```
// routes/events.ts --- stats endpoint backed by the view
const rows = await sql`
    SELECT event_id, title, status, capacity, tickets_sold, spots_remaining
    FROM v_event_stats
    WHERE status != 'cancelled' AND end_time >= ${now}
`;
```

3.6. Stored Function

`check_event_capacity(p_event_id INTEGER)` is a PL/pgSQL function that returns a single row containing the event’s capacity, current ticket count, spots remaining, and an `is_available` boolean. It is declared `STABLE` because it reads but does not modify data. The function is called inside the ticket purchase transaction — after the `SELECT ... FOR UPDATE` row lock is acquired — to atomically determine whether the event can still accept a registration.

```

CREATE OR REPLACE FUNCTION check_event_capacity(p_event_id INTEGER)
RETURNS TABLE (
    event_id      INTEGER,
    title         VARCHAR,
    capacity      INTEGER,
    tickets_sold  BIGINT,
    spots_remaining BIGINT,
    is_available  BOOLEAN
) LANGUAGE plpgsql STABLE AS $$
BEGIN
    RETURN QUERY
        SELECT e.id, e.title, e.capacity,
               COUNT(t.id),
               (e.capacity - COUNT(t.id)),
               (e.capacity - COUNT(t.id)) > 0
        FROM events e
        LEFT JOIN tickets t ON e.id = t.event_id
        WHERE e.id = p_event_id
        GROUP BY e.id, e.title, e.capacity;
END;
$$;

```

4. Main Functionalities

4.1. Student

Students are the primary users of the platform. After registering with a university email and NetID, a student can:

- **Browse events** — Search by keyword (case-insensitive `ILIKE`), filter by category, filter by date range, and sort by date or title. All event statuses are visible.
- **View event detail** — See the full event description, venue, time range, ticket price, and real-time remaining capacity computed by `check_event_capacity`.
- **Book a ticket** — Click “Book Ticket” on any upcoming event with capacity remaining. The backend issues a `SELECT ... FOR UPDATE + INSERT` transaction and returns a unique confirmation code.
- **Manage bookings** — The My Tickets page lists all bookings with confirmation codes and check-in status. Students can cancel a ticket or self-check-in during the 30-minute window before the event starts through its end time.
- **Edit profile** — Update display name, email, and upload a profile photo stored in the `images` table.

4.2. Organizer

Organizers create and manage events. The Organizer Dashboard provides:

- **Stats overview** — Stat cards show total events created, upcoming event count, and total attendees checked in, drawn from `v_event_stats`.

- **Create an event** — A full creation form collects title, description, location, time range, capacity, ticket price, and one or more categories. An optional Mapbox GL JS map picker lets the organizer drop a GPS pin, which renders as an interactive map preview on the event detail page. A banner image can be uploaded.
- **Edit events** — Any field of an owned event can be updated; changes issue a dynamic `UPDATE` with only the changed fields.
- **Cancel an event** — Sets `status = 'cancelled'` via a targeted `UPDATE`.
- **View attendees** — The attendee panel lists all ticket holders with name, email, check-in status, and confirmation code.

4.3. Administrator

Admins have unrestricted access via the Admin Panel:

- **User management** — View all registered accounts. Promote or demote any user's role in real time via an inline dropdown (issues `PATCH /api/users/:id/role`).
- **Event management** — Full CRUD over all events in the system, regardless of organizer ownership.
- **Category management** — Create and delete event categories.
- **All organizer capabilities** — Admins can create and manage events on behalf of any organizer.

5. Interface Pages

The frontend ships eight routed pages, well above the three-page minimum. Routing is managed by `react-router-dom` v7; protected routes redirect unauthenticated users to `/login`. The UI uses a glassmorphism design system with a deep-green brand color, and supports full light and dark mode.

Page (in <code>frontend/src/pages/</code>)	Route	Purpose
<code>Landing.tsx</code>	<code>/</code>	Public landing page with live event count and feature summary
<code>Login.tsx</code>	<code>/login</code>	JWT-based sign-in form
<code>Register.tsx</code>	<code>/register</code>	Account creation; role defaults to <code>student</code>
<code>BrowseEvents.tsx</code>	<code>/events</code>	Event discovery: keyword search, category filter, date range, status filter
<code>EventDetail.tsx</code>	<code>/events/:id</code>	Full event info, real-time remaining capacity, Book Ticket action
<code>MyTickets.tsx</code>	<code>/my-tickets</code>	Student's active bookings with confirmation codes and check-in control
<code>OrganizerDashboard.tsx</code>	<code>/dashboard</code>	Create, edit, and cancel events; attendee list; ticket stats

Page (in frontend/src/pages/)	Route	Purpose
AdminPanel.tsx	/admin	System management: users, events, categories, role assignment

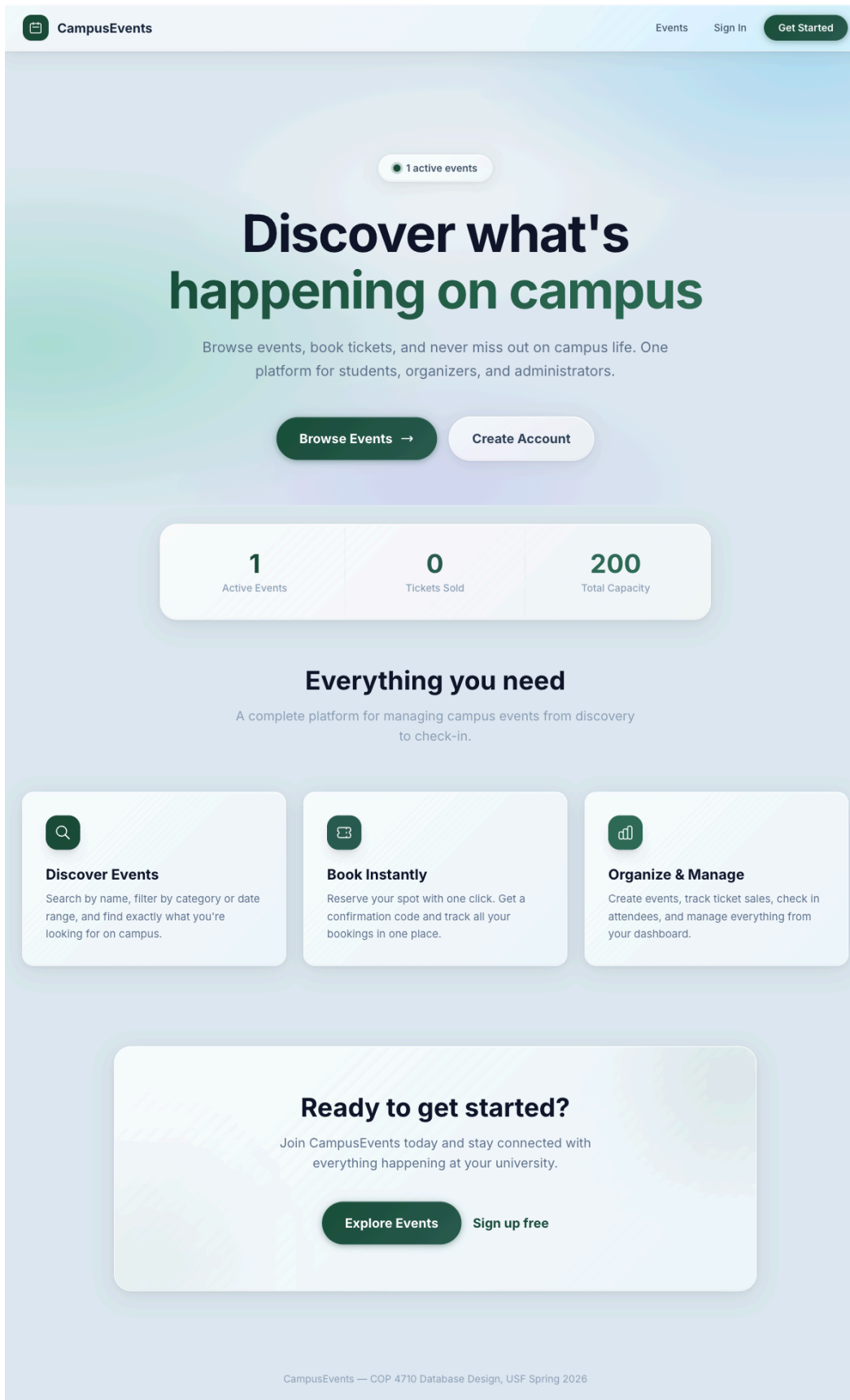


Figure 3: **Landing (/)** — public hero section. The active-event count and total capacity figures are fetched live from `GET /api/events` on every page load, pulling real data from the database.

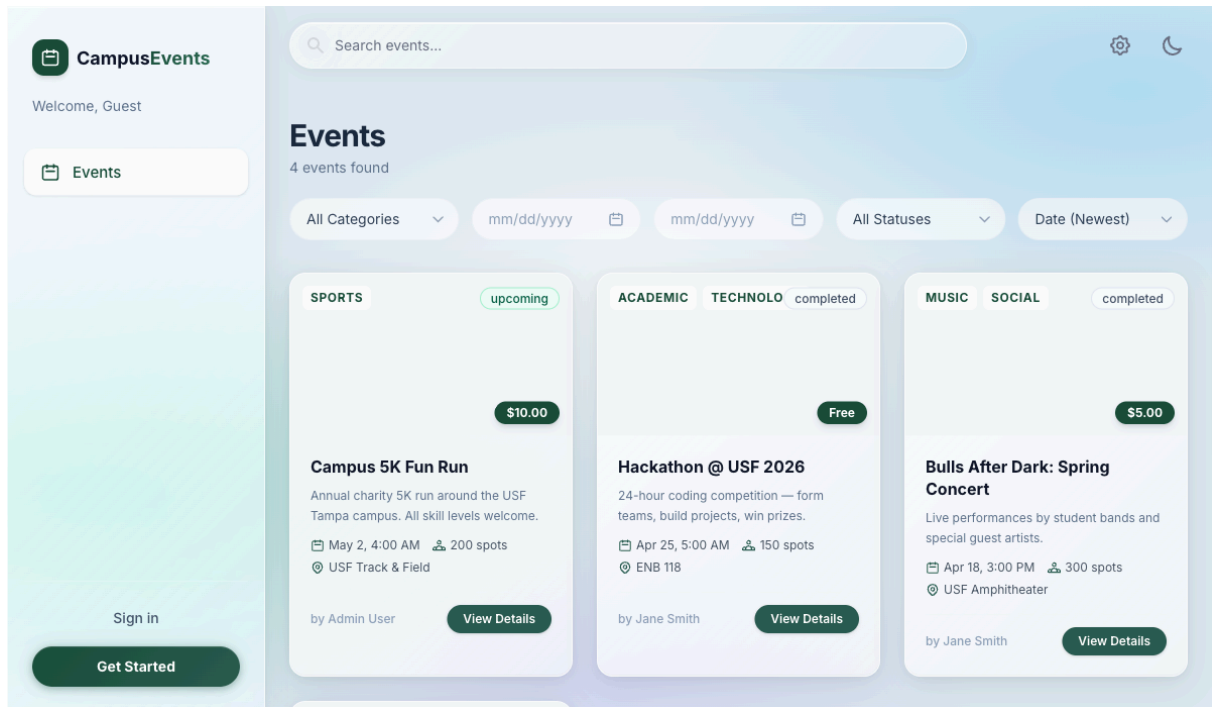


Figure 4: **Browse Events** (/events) — event grid backed by `GET /api/events`. Keyword search uses `ILIKE`, category filter JOINS through `EVENT_CATEGORIES`, date range uses `BETWEEN`, and status filter applies a `WHERE` clause — all composed server-side.

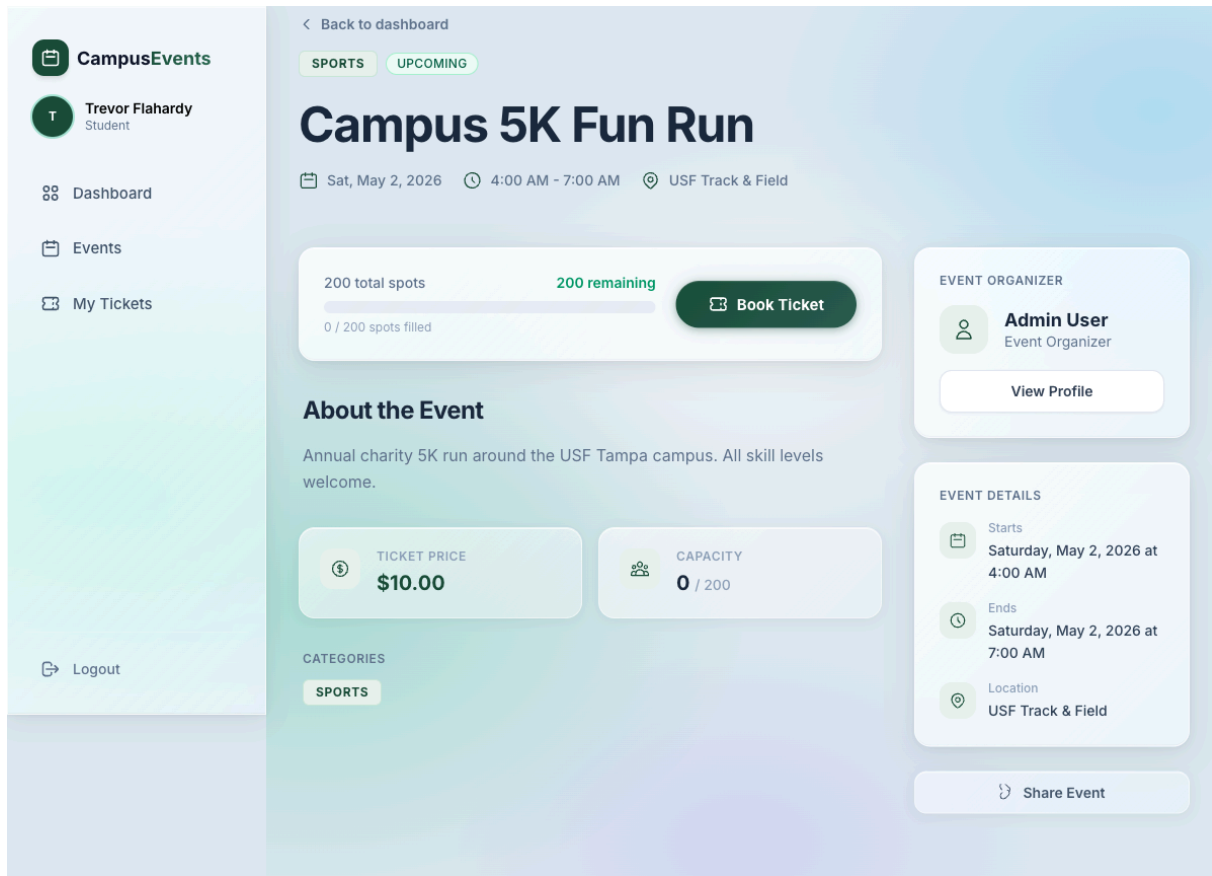


Figure 5: **Event Detail** (`/events/:id`) — full event info with a real-time remaining-capacity counter. Capacity is computed by the `check_event_capacity` stored function. The Book Ticket button triggers the `SELECT ... FOR UPDATE + INSERT` transaction.

CampusEvents

Trevor Flahardy
Student

Dashboard

Events

My Tickets

Logout

My Tickets

View and manage your upcoming campus experiences. Your digital entry passes for the season.

CAREER completed

Spring Career Fair 2026

Marshall Student Center Ballroom

Event ended

DATE	TIME	PRICE	CHECK-IN
Apr 10, 2026	6:00 AM	Free	No

VERIFICATION

CONF. NUMBER

CF26A001

[View Event](#)

ACADEMIC TECHNOLOGY completed

Hackathon @ USF 2026

ENB 118

Event ended

DATE	TIME	PRICE	CHECK-IN
Apr 25, 2026	5:00 AM	Free	No

VERIFICATION

CONF. NUMBER

HACK2601

[View Event](#)

Showing 2 active tickets

Figure 6: **My Tickets** (/my-tickets) — authenticated student's bookings. Each card shows the unique confirmation code, check-in status, and full event info joined across TICKETS, EVENTS, and CATEGORIES.

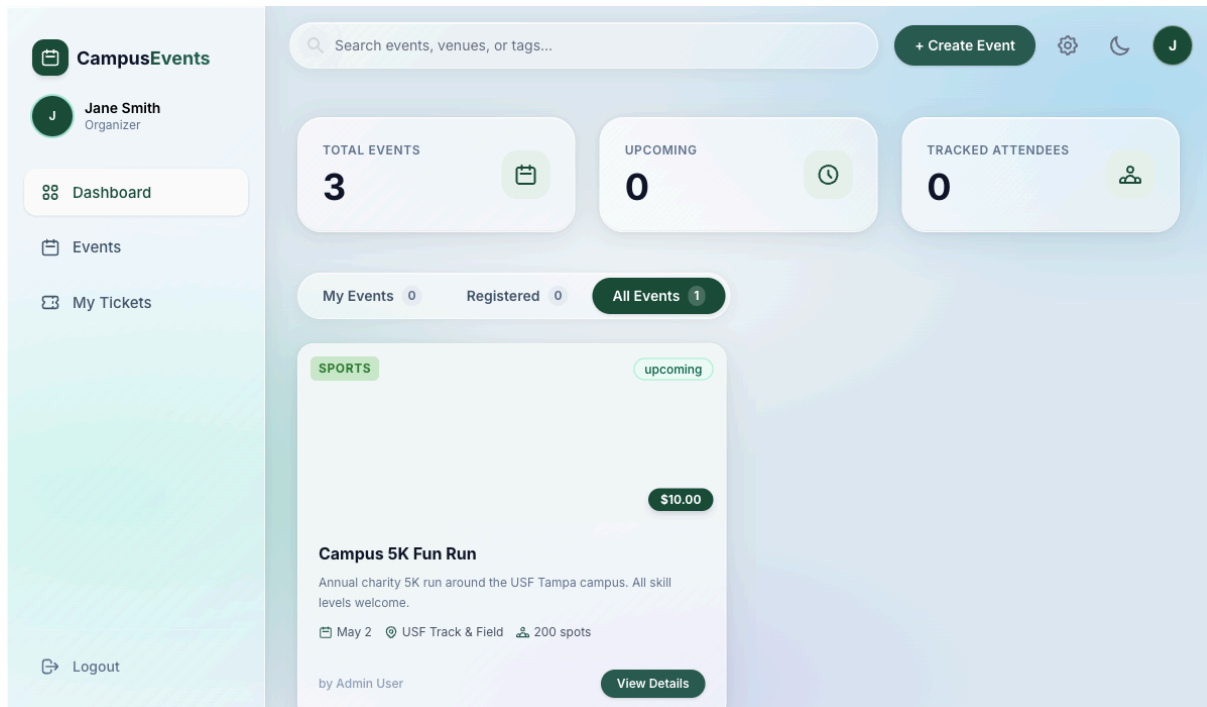


Figure 7: **Organizer Dashboard** (/dashboard) — stat cards and event grid. Ticket totals come from the `v_event_stats` view. The All Events tab shows every event in the system; the “+ Create Event” button opens a full creation form with an optional Mapbox map picker.

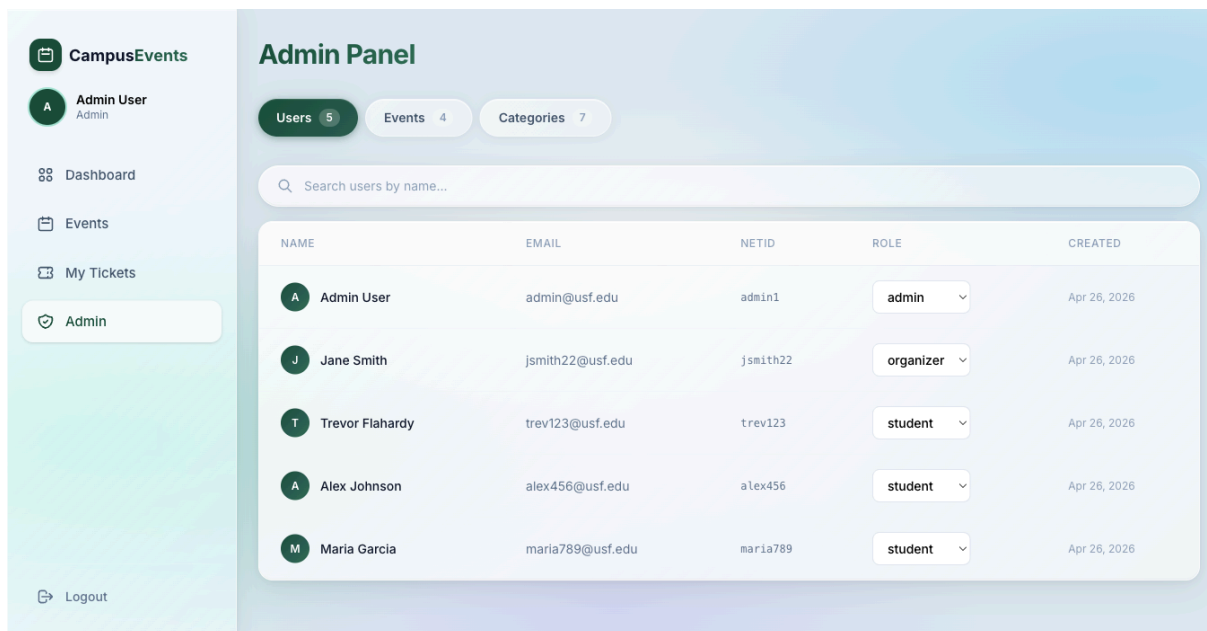


Figure 8: **Admin Panel** (/admin) — full user directory with inline role dropdowns. Selecting a new role immediately issues `PATCH /api/users/:id/role`; the change takes effect on the target user’s next login. The Events and Categories tabs provide full system-wide CRUD.

6. SQL Query Implementation

All ten distinct SQL query types are implemented as live REST endpoints driving real UI features. Every query is issued through `postgres.js` tagged template literals, which parameterize all user-supplied values automatically. The table below maps each query type to its handler file and the page that exercises it.

#	Query Type	Endpoint (handler)	UI Caller
Q1	SELECT + JOIN	GET /api/events routes/events.ts	BrowseEvents.tsx
Q2	GROUP BY + COUNT	GET /api/events/stats routes/events.ts	OrganizerDashboard.tsx
Q3	SELECT + subquery	GET /api/events/:id routes/events.ts	EventDetail.tsx
Q4	SELECT + multi-JOIN	GET /api/tickets/ user/:id routes/tickets.ts	MyTickets.tsx
Q5	INSERT (transaction)	POST /api/tickets routes/tickets.ts	EventDetail.tsx
Q6	UPDATE	PATCH /api/tickets/:id/ checkin routes/tickets.ts	MyTickets.tsx
Q7	UPDATE	PATCH /api/events/:id routes/events.ts	OrganizerDashboard.tsx
Q8	DELETE	DELETE /api/tickets/:id routes/tickets.ts	MyTickets.tsx
Q9	SELECT + HAVING	GET /api/categories/ popular routes/categories.ts	BrowseEvents.tsx
Q10	SELECT + BETWEEN	GET /api/events? from=&to= routes/events.ts	BrowseEvents.tsx

6.1. Q1 + Q10: Events List with JOIN and Date Filtering

The browse endpoint dynamically composes a `WHERE` clause from optional query parameters. Each condition is a safe tagged-template fragment; they are reduced with `AND` to form the full predicate.

```
// routes/events.ts --- GET /api/events
const conditions = [sql`TRUE`];
// Q10: BETWEEN-style date filtering via >= / <=
if (from) conditions.push(sql`e.start_time >= ${new Date(from)}`);
if (to) conditions.push(sql`e.start_time <= ${new Date(to)}`);
if (search) conditions.push(sql`e.title ILIKE ${"%" + search + "%"}`);
const where = conditions.reduce((a, c) => sql`${a} AND ${c}`);

// Q1: SELECT with JOIN --- events with organizer name
const rows = await sql`
  SELECT e.id, e.title, e.description, e.location,
         e.start_time, e.end_time, e.capacity, e.ticket_price,
         e.status, e.banner_url, e.latitude, e.longitude,
         u.name AS organizer_name
  FROM events e
  INNER JOIN users u ON e.organizer_id = u.id
  WHERE ${where}
  ORDER BY e.start_time
`;
```

When a category filter is active, a second `INNER JOIN` through `event_categories` restricts results to that category.

6.2. Q2: Ticket Sales Aggregation (via View)

The stats endpoint queries `v_event_stats`, which pre-aggregates `tickets_sold` and `spots_remaining` using `GROUP BY + COUNT` on the underlying `EVENTS-TICKETS` join. Using the view keeps the route lean and demonstrates the database-side view bonus feature.

```
// routes/events.ts --- GET /api/events/stats
const rows = await sql`
  SELECT event_id, title, status, capacity, tickets_sold, spots_remaining
  FROM v_event_stats
  WHERE status != 'cancelled' AND end_time >= ${now}
`;
```

6.3. Q3: Remaining Capacity via Correlated Subquery

The event detail endpoint computes `spots_remaining` inline using a correlated subquery against the `TICKETS` table. This gives an accurate real-time count without a separate query round-trip.

```
// routes/events.ts --- GET /api/events/:id
const rows = await sql`
  SELECT e.id, e.title, e.description, e.location,
         e.start_time, e.end_time, e.capacity, e.ticket_price,
         e.status, e.banner_url, e.latitude, e.longitude,
         u.name AS organizer_name,
         (e.capacity - (SELECT COUNT(*)::int FROM tickets WHERE event_id =
```

```

e.id))
    AS spots_remaining
FROM events e
INNER JOIN users u ON e.organizer_id = u.id
WHERE e.id = ${id}
`;

```

6.4. Q4: Multi-JOIN Ticket History

The My Tickets endpoint joins TICKETS with EVENTS in one query, then batch-fetches all associated categories in a second query using `ANY(array)` to avoid an N+1 pattern. Results are merged in application code before returning.

```

// routes/tickets.ts --- GET /api/tickets/user/:userId
const rows = await sql`
  SELECT t.id AS ticket_id, t.purchased_at, t.checked_in, t.confirmation_code,
         e.id AS event_id, e.title AS event_title, e.location AS
event_location,
         e.start_time AS event_start_time, e.end_time AS event_end_time,
         e.status AS event_status, e.ticket_price
  FROM tickets t
  INNER JOIN events e ON t.event_id = e.id
  WHERE t.user_id = ${userId}
  ORDER BY e.start_time
`;

// Batch-fetch categories for all event IDs in one shot (avoids N+1)
const categoryRows = await sql`
  SELECT ec.event_id, c.id, c.name
  FROM categories c
  INNER JOIN event_categories ec ON c.id = ec.category_id
  WHERE ec.event_id = ANY(${eventIds})
`;

```

6.5. Q5: Ticket Purchase (INSERT with Transaction + Stored Function)

The ticket purchase flow wraps the capacity check and INSERT in a single database transaction with a row-level lock on the event. The `SELECT ... FOR UPDATE` blocks any concurrent purchaser until this transaction commits, so the ticket count read by `check_event_capacity` is authoritative at insert time — the mechanism that actually prevents oversell under concurrent requests.

```

// routes/tickets.ts --- POST /api/tickets
const outcome = await sql.begin(async (tx) => {
  // Lock the event row to serialize concurrent purchase attempts
  const [event] = await tx`
    SELECT id, capacity, status FROM events WHERE id = ${eventId} FOR UPDATE
  `;

```

```

    if (!event) return { kind: "not_found" };
    if (event.status === "cancelled") return { kind: "cancelled" };

    // Stored function checks current capacity within the held lock
    const [cap] = await tx`SELECT * FROM check_event_capacity(${eventId})`;
    if (!cap || !cap.isAvailable) return { kind: "sold_out" };

    // Q5: INSERT new ticket and return the created row
    const [inserted] = await tx`
      INSERT INTO tickets (user_id, event_id, confirmation_code)
      VALUES (${userId}, ${eventId}, ${confirmationCode})
      RETURNING *
    `;
    return { kind: "ok", ticket: inserted };
  });

```

6.6. Q6–Q8: State-Modifying Queries

```

// Q6: UPDATE --- check in a ticket (toggle boolean flag)
const [updated] = await sql`
  UPDATE tickets SET checked_in = TRUE WHERE id = ${id} RETURNING *
`;

// Q7: UPDATE --- cancel or edit an event (dynamic SET clause)
// sql() helper maps a plain object's keys to SQL column names safely
const [updated] = await sql`
  UPDATE events
  SET ${sql(updates, ...Object.keys(updates))}
  WHERE id = ${id}
  RETURNING *
`;

// Q8: DELETE --- cancel (remove) a ticket
const deleted = await sql`DELETE FROM tickets WHERE id = ${id} RETURNING *`;

```

6.7. Q9: Popular Categories (HAVING)

```

// routes/categories.ts --- GET /api/categories/popular
const rows = await sql`
  SELECT c.id AS category_id, c.name,
         COUNT(ec.event_id)::int AS event_count
  FROM categories c
  INNER JOIN event_categories ec ON c.id = ec.category_id
  GROUP BY c.id, c.name
  HAVING COUNT(ec.event_id) > 1
`;

```

7. Authentication & Authorization

User accounts with `userID`/password are implemented as bonus feature 1. Passwords are hashed with `bcrypt` (`@node-rs/bcrypt`, cost factor 10) on registration and verified on login. A successful login returns a JWT signed with `HS256` containing the user's `id` and `role`. The token is stored in `localStorage` on the client and attached to subsequent requests via `Authorization: Bearer`.

All protected routes pass through `authMiddleware` at `backend/src/middleware/auth.ts`, which verifies the token and injects `userId` and `userRole` into the request context. A `requireRole(...)` factory then gates individual routes to specific roles.

```
// middleware/auth.ts
export async function authMiddleware(c: Context<AuthEnv>, next: Next) {
  const header = c.req.header("Authorization");
  if (!header?.startsWith("Bearer "))
    return c.json({ error: "Missing or invalid Authorization header" }, 401);
  const payload = await verify(header.slice(7), JWT_SECRET, "HS256");
  c.set("userId", payload.sub as number);
  c.set("userRole", payload.role as string);
  await next();
}

export function requireRole(...roles: string[]) {
  return async (c: Context<AuthEnv>, next: Next) => {
    if (!roles.includes(c.get("userRole")))
      return c.json({ error: "Forbidden" }, 403);
    await next();
  };
}
```

Role enforcement is applied at the route level: `POST /api/events` requires `organizer` or `admin`; `PATCH /api/users/:id/role` and `GET /api/users` require `admin`; organizers may only modify their own events (ownership verified with `SELECT WHERE organizer_id = $userId` before any mutation). Every write endpoint additionally validates its request body with a Zod schema before any SQL executes, so the database never receives malformed input.

8. How to Run

The entire stack boots with one command from the project root:

```
docker compose up
```

Docker Compose starts three containers — `campusevents_db` (PostgreSQL 16), `campusevents_api` (Bun + Hono), and `campusevents_web` (React + Vite). On startup, the

backend applies `schema.sql` via `postgres.js`'s `unsafe()` method, creating all six tables, five indexes, two views, and the stored function. All DDL statements use `IF NOT EXISTS` or `CREATE OR REPLACE`, making the migration idempotent and safe to run on every boot. The backend logs  `Schema applied` when ready; Vite logs the local URL when the bundle is served.

```
# Load demo users, events, and tickets
cd backend && bun run db:seed

# Open the app
open http://localhost:5173

# Backend health check
curl http://localhost:3000/health # → {"status":"ok"}
```

Email	Password	Role
admin@usf.edu	password123	Administrator — full system access
jsmith22@usf.edu	password123	Organizer — event creation and management
trev123@usf.edu	password123	Student — browse, register, check in

```
futur@TrevMacbook project_4 % docker compose up --build
campusevents_api | message: "relation \"idx_tickets_event\" already exists, skipping",
campusevents_api | file: "index.c",
campusevents_api | line: "877",
campusevents_api | routine: "index_create",
campusevents_api | }
campusevents_api | ✓ Schema applied
campusevents_api | ⚡ Server running on http://localhost:3000
campusevents_api | Started development server: http://localhost:3000
campusevents_web | $ vite --host "0.0.0.0"
campusevents_web |
campusevents_web | VITE v8.0.2 ready in 294 ms
campusevents_web |
campusevents_web | → Local: http://localhost:5173/
campusevents_web | → Network: http://172.21.0.4:5173/
campusevents_api | <-- GET /api/auth/me
campusevents_api | --> GET /api/auth/me 200 12ms
campusevents_api | <-- GET /api/auth/me
campusevents_api | --> GET /api/auth/me 200 1ms
campusevents_api | <-- GET /api/events
campusevents_api | <-- GET /api/users
campusevents_api | <-- GET /api/categories
campusevents_api | <-- GET /api/categories/popular
campusevents_api | --> GET /api/events 200 5ms
campusevents_api | <-- GET /api/events
campusevents_api | --> GET /api/categories/popular 200 8ms
campusevents_api | --> GET /api/categories 200 8ms
campusevents_api | --> GET /api/users 200 10ms
campusevents_api | --> GET /api/events 200 1ms
campusevents_api | <-- GET /api/categories
campusevents_api | <-- GET /api/categories/popular
campusevents_api | <-- GET /api/users
campusevents_api | --> GET /api/categories 200 1ms
campusevents_api | --> GET /api/users 200 0ms
campusevents_api | --> GET /api/categories/popular 200 2ms

View in Docker Desktop View Config Enable Watch
```

Figure 9: `docker compose up` — all three tiers starting in sequence. The API logs “Schema applied” before accepting traffic; Vite serves the React bundle on port 5173. The seed script can be run immediately after to populate demo data.

```
o futur@TrevMacbook project_4 % docker exec -it campusevents_db psql -U campus -d campusevents
psql (16.13)
Type "help" for help.

campusevents=# \dt
List of relations
Schema | Name | Type | Owner
-----+-----+-----+-----
public | categories | table | campus
public | event_categories | table | campus
public | events | table | campus
public | images | table | campus
public | tickets | table | campus
public | users | table | campus
(6 rows)

campusevents=#
```

Figure 10: Output of `\dt` in `psql` against the running container — six live relations created by `schema.sql`. Running `\dv` additionally shows `v_event_stats` and `v_upcoming_events`; `\df` shows `check_event_capacity`.

9. Status Against Course Requirements

The table below maps every graded requirement and bonus item from the assignment handout to our implementation status.

#	Requirement	Status
1	DBMS-backed database with ≥ 3 relations loaded with data	✓ 6 tables + seed
2	≥ 8 distinct SQL query types including state-modifying queries	✓ 10 implemented
3	Web-based interface with ≥ 3 distinct pages	✓ 8 pages
4	JDBC/ODBC-style connection to send queries to the database	✓ postgres.js raw SQL
—	Bonus — user accounts with userID/password	✓ bcrypt + JWT
—	Bonus — views with per-role projections	✓ 2 views in schema.sql
—	Bonus — stored procedures/functions on the database side	✓ check_event_capacity()
—	Bonus — client-side scripts for application logic	✓ React 19 SPA
—	Bonus — additional relevant features	✓ transactions, image upload, Mapbox maps, dark mode

All four required features are fully implemented. For the bonus items: user accounts use bcrypt password hashing and JWT-based session tokens; two SQL views (`v_upcoming_events` , `v_event_stats`) provide role-scoped projections of the underlying tables; the `check_event_capacity` PL/pgSQL function encapsulates capacity logic on the database side and is called from within the ticket purchase transaction; and the React single-page application handles all client-side routing, state management, and UI rendering. Additional features beyond the rubric include a row-level-locked ticket purchase transaction to prevent oversell, base-64 image storage in the database for media persistence, Mapbox GL JS map previews for event venues, and full light/dark mode theming.